2021 Ph H2 Q11

Section: Particles and Waves

Topic: Spectra

This question explores how a star's spectrum can be used to estimate temperature.

(a)
Estimate the surface temperature of the star

Use **Wien's Law**:

$$\lambda_{\max} T = 2.9 \times 10^{-3}\,\mathrm{m\,K} \quad \Rightarrow \quad T = \frac{2.9 \times 10^{-3}}{\lambda_{\max}}$$

Given:

- $\lambda_{\max} = 4.8 \times 10^{-7}\,\mathrm{m}$

$$T = \frac{2.9 \times 10^{-3}}{4.8 \times 10^{-7}} = 6.0 \times 10^{3}\,\mathrm{K}$$

(b)
Explain why the spectrum of the star does not show a continuous range of wavelengths

✅ The star's spectrum is not continuous because **certain wavelengths of light are absorbed** by elements in the outer layers of the star.
These elements absorb specific frequencies of radiation, producing **absorption lines** in the spectrum.

(c)
Explain how absorption lines can be used to identify elements in the star's atmosphere

✅ Every element has a **unique set of energy levels**, and therefore absorbs **specific wavelengths** of light.
The **pattern of absorption lines** can be matched to known spectral fingerprints of elements on Earth, allowing identification of elements in the star's atmosphere.

🧠 Revision Tips

- Use **Wien's Law** when given peak wavelength to estimate surface temperature
- Absorption lines are **missing wavelengths** in the continuous spectrum
- These lines result from **electrons absorbing energy** and jumping to higher energy levels
- Comparing absorption patterns is how we determine **stellar compositi**